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Doelgroep: medestudnenten

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Sabawoon Enayat

20230497

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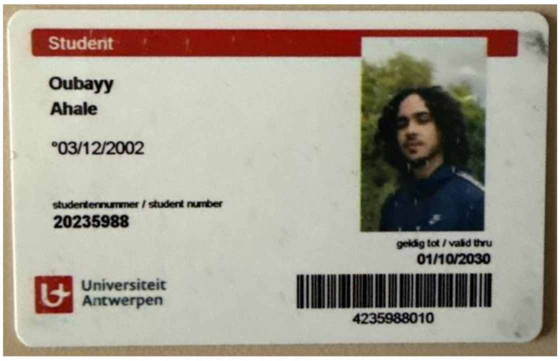

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(studentenkaart is nog niet aangekomen)

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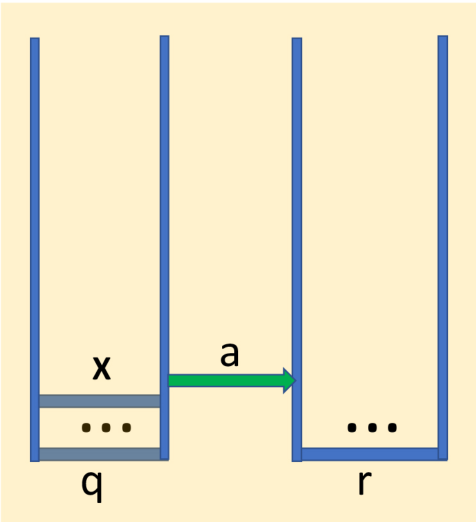
Toelichting constructie van PDA naar CFG

Wat betekenen die nieuwe regels?

Formally, let $P = (Q, \Sigma, \Gamma, \delta, q_0, Z_0)$ be a PDA. Define $G = (V, \Sigma, R, S)$, where

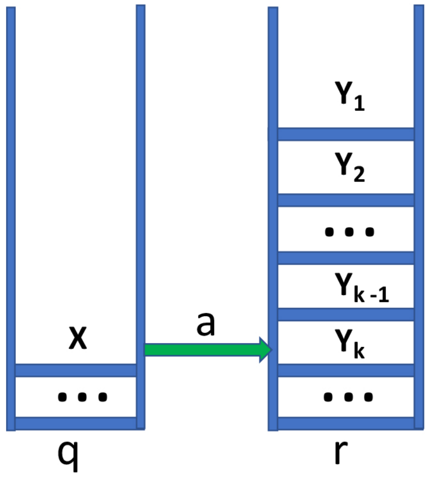
$$
\begin{aligned}
V &= \{[pXq] : \{p, q\} \subseteq Q, X \in \Gamma\} \cup \{S\} \\
R &= \{S \to [q_0 Z_0 p] : p \in Q\} \cup \\
&\quad \{\underline{[\mathbf{q} X r_k] \to a[\mathbf{r} Y_1 r_1][\mathbf{r}_1 Y_2 r_2] \cdots [r_{k-1} Y_k r_k]} : \\
&\qquad\qquad a \in \Sigma \cup \{\epsilon\}, \\
&\qquad\qquad \{r_1, \ldots, r_k\} \subseteq Q, \\
&\qquad\qquad (\mathbf{r}, Y_1 Y_2 \cdots Y_k) \in \delta(\mathbf{q}, a, X)\} \\
&\quad \cup \{\underline{[\mathbf{q} X r] \to a : a \in \Sigma \cup \{\epsilon\}}, \\
&\qquad\qquad (\mathbf{r}, \epsilon) \in \delta(\mathbf{q}, a, X)\}
\end{aligned}
$$

$$(\mathbf{r}, \epsilon) \in \delta(\mathbf{q}, a, X)$$

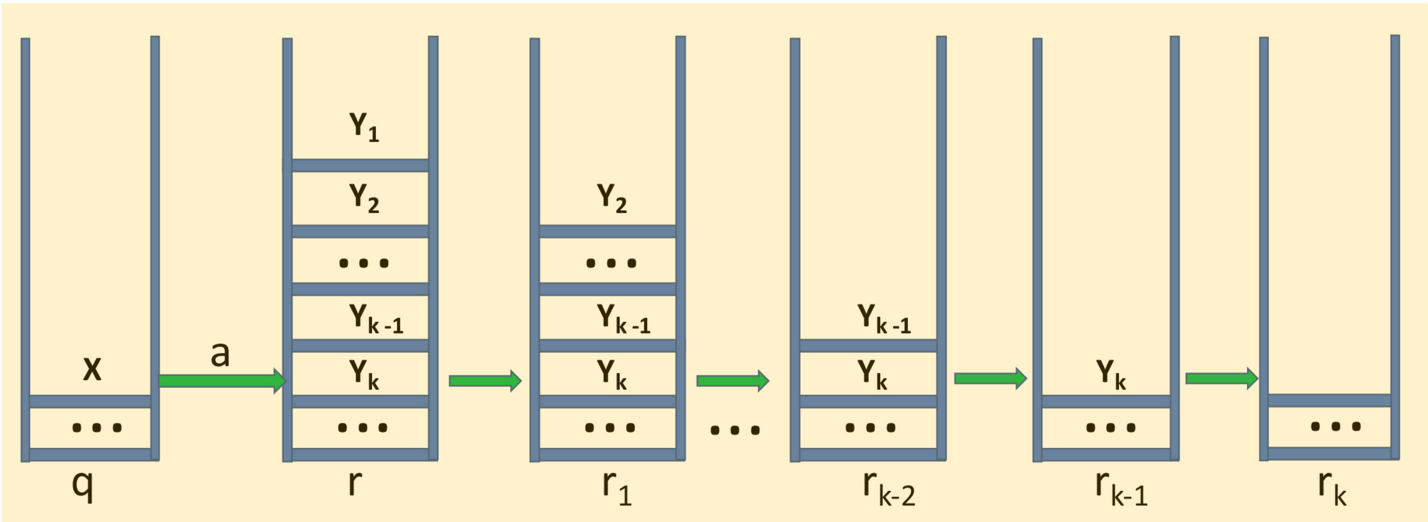


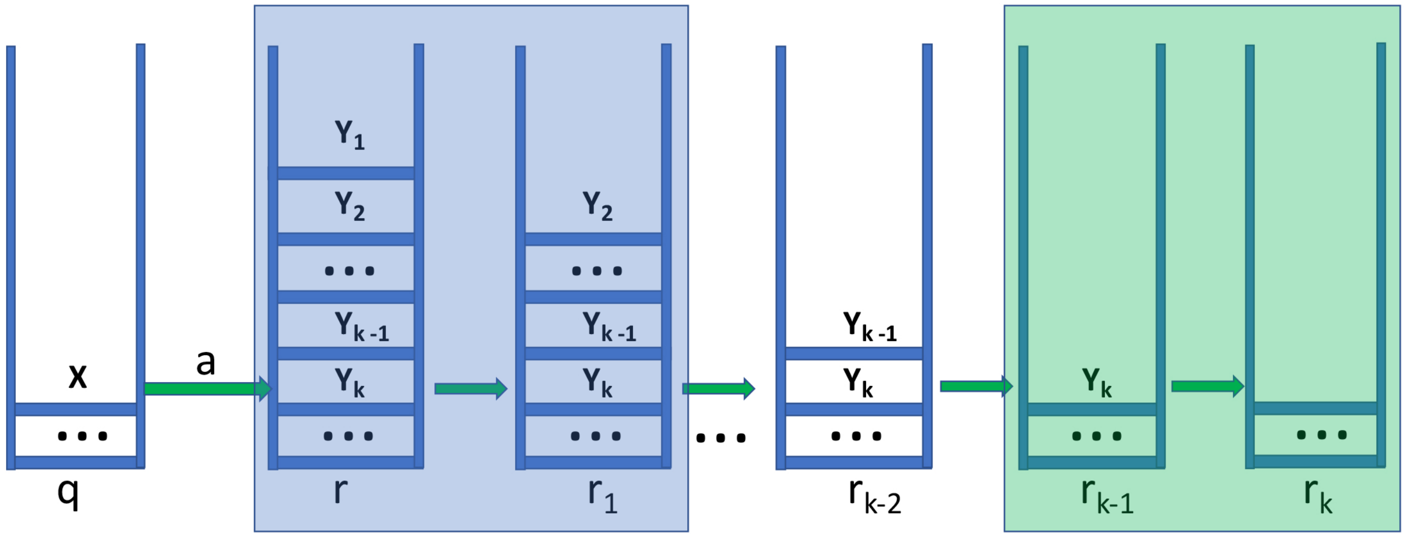
$$[\mathbf{q}Xr] \rightarrow a$$

$$(\mathbf{r}, Y_1 Y_2 \cdots Y_k) \in \delta(\mathbf{q}, a, X)$$



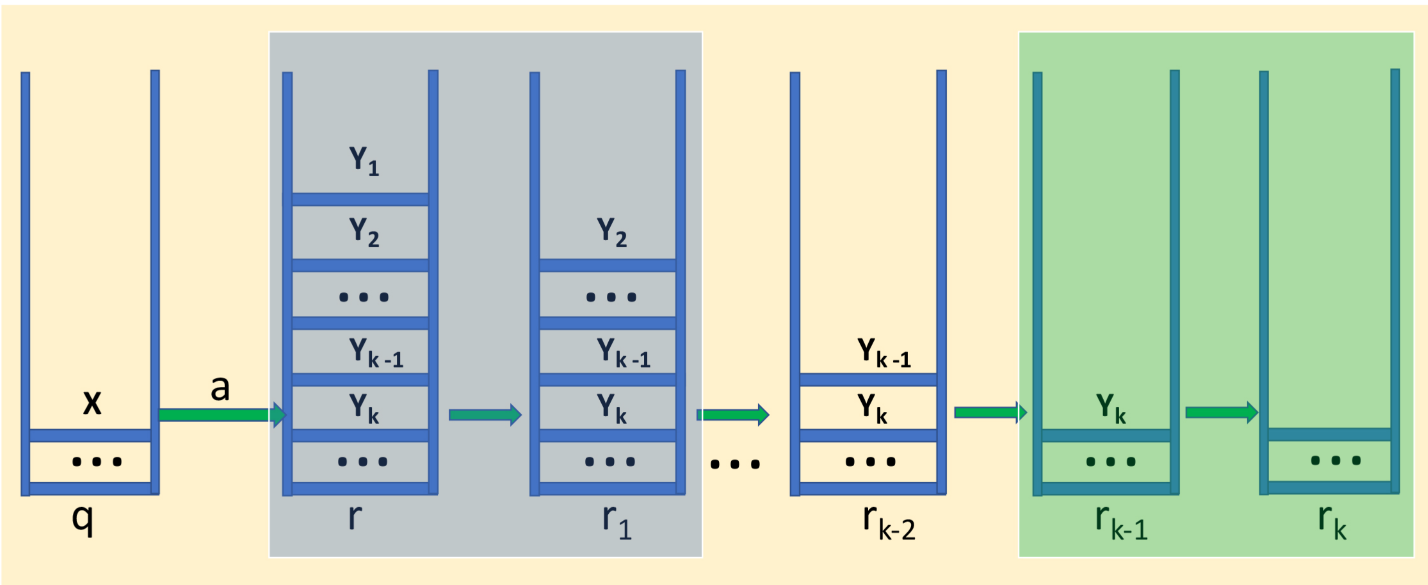
$$[\mathbf{q} X r_k]$$





$$[\mathbf{r}Y_1r_1]\cdots[r_{k-1}Y_kr_k]$$

$$[\mathbf{q}Xr_k] \rightarrow a[\mathbf{r}Y_1r_1]\cdots[r_{k-1}Y_kr_k]$$



Theorem 6.14: Let G be constructed from a PDA P as above. Then $L(G) = N(P)$

Proof:

($\supseteq$-*direction.*) We shall show by an induction on the length of the sequence $\overset{*}{\vdash}$ that

(♠) If $\underbrace{(q, w, X) \overset{*}{\vdash} (p, \epsilon, \epsilon)}_{w \in N(P)}$ then $\underbrace{[qXp] \overset{*}{\Rightarrow} w}_{w \in L(G)}$.

$w \in N(P) \quad \Rightarrow \quad w \in L(G)$

Basis: Length 1. Then w is an a or ϵ, and $(p, \epsilon) \in \delta(q, w, X)$. By the construction of G we have $[qXp] \rightarrow w$ and thus $[qXp] \overset{*}{\Rightarrow} w$.

Inductie hypothese:
We stellen dat (♠) geldt voor lengtes < n, dan tonen we dat het geldt voor lengte n.

Induction: Length is $n > 1$, and ♠ holds for lengths $< n$. We must have a $\vdash$ sequence

$$(q, w, X) \vdash (r_0, x, Y_1 Y_2 \cdots Y_k) \vdash \cdots \vdash (p, \epsilon, \epsilon),$$

where $w = ax$ and $a \in \Sigma \cup \{\epsilon\}$. It follows that $(r_0, Y_1 Y_2 \cdots Y_k) \in \delta(q, a, X)$. Then we have a production

$$[qXr_k] \to a[r_0 Y_1 r_1] \cdots [r_{k-1} Y_k r_k],$$

for all $\{r_1, \ldots, r_{k-1}\} \subseteq Q$ and $r_k = p$.
We may choose r_i to be the state of the PDA in the above $\vdash$ sequence when Y_i is popped. Let $w = aw_1 w_2 \cdots w_k$, where w_i is consumed while Y_i is popped. Then

$$(r_{i-1}, w_i, Y_i) \overset{*}{\vdash} (r_i, \epsilon, \epsilon).$$

By the IH we get

$$[r_{i-1} Y_i r_i] \overset{*}{\Rightarrow} w_i$$

We then get the following derivation sequence:

$$[qXr_k] \Rightarrow a[r_0Y_1r_1]\cdots[r_{k-1}Y_kr_k] \stackrel{*}{\Rightarrow}$$

$$aw_1[r_1Y_2r_2][r_2Y_3r_3]\cdots[r_{k-1}Y_kr_k] \stackrel{*}{\Rightarrow}$$

$$aw_1w_2[r_2Y_3r_3]\cdots[r_{k-1}Y_kr_k] \stackrel{*}{\Rightarrow}$$

$$\cdots$$

$$aw_1w_2\cdots w_k = w$$

and $r_k = p$.

($\subseteq$-*direction.*) We shall show by an induction on the length of the derivation $\stackrel{*}{\Rightarrow}$ that

($\heartsuit$) If $[qXp] \stackrel{*}{\Rightarrow} w$ then $(q, w, X) \stackrel{*}{\vdash} (p, \epsilon, \epsilon)$

$w \in L(G) \quad \Rightarrow \quad w \in N(P)$

Basis: One step. Then we have a production $[qXp] \rightarrow w$. From the construction of G it follows that $(p, \epsilon) \in \delta(q, a, X)$, where $w = a$. But then $(q, w, X) \stackrel{*}{\vdash} (p, \epsilon, \epsilon)$.

Inductie hypothese:
We stellen dat ($\heartsuit$) geldt voor lengtes < n, dan tonen we dat het geldt voor lengte n.

Induction: Length of $\stackrel{*}{\Rightarrow}$ is $n > 1$, and $\heartsuit$ holds for lengths $< n$. Then we must have

$$[qXr_k] \Rightarrow a[r_0Y_1r_1][r_1Y_2r_2]\cdots[r_{k-1}Y_kr_k] \stackrel{*}{\Rightarrow} w$$

where $r_k = p$. This must come from the transition $(r_0, Y_1Y_2\cdots Y_k) \in \delta(q, a, X)$ and consequently

$$(q, aw_1w_2\cdots w_k, X) \vdash (r_0, w_1w_2\cdots w_k, Y_1Y_2\cdots Y_k)$$

We can break w into $aw_1w_2\cdots w_k$ such that $[r_{i-1}Y_ir_i] \stackrel{*}{\Rightarrow} w_i$. From the IH we get

$$(r_{i-1}, w_i, Y_i) \stackrel{*}{\vdash} (r_i, \epsilon, \epsilon)$$

From Theorem 6.5 we get
$(r_{i-1}, w_iw_{i+1}\cdots w_k, Y_iY_{i+1}\cdots Y_k) \stackrel{*}{\vdash}$
$(r_i, w_{i+1}\cdots w_k, Y_{i+1}\cdots Y_k)$

Since this holds for all $i \in \{1, \ldots, k\}$, we get
$(q, aw_1w_2\cdots w_k, X) \vdash$
$(r_0, w_1w_2\cdots w_k, Y_1Y_2\cdots Y_k) \stackrel{*}{\vdash}$
$(r_1, w_2\cdots w_k, Y_2\cdots Y_k) \stackrel{*}{\vdash}$
$(r_2, w_3\cdots w_k, Y_3\cdots Y_k) \stackrel{*}{\vdash}$
$\cdots$
(p, ϵ, ϵ).

So we have shown
$(q, w, X) \overset{*}{\vdash} (p, \epsilon, \epsilon)$ if and only if $[qXp] \overset{*}{\Rightarrow} w$.

In particular for $q = q_0$ and $X = Z_0$
$(q_0, w, Z_0) \overset{*}{\vdash} (p, \epsilon, \epsilon)$ if and only if $[q_0 Z_0 p] \overset{*}{\Rightarrow} w$

and because of the rule $S \rightarrow [q_0 Z_0 p]$
$[q_0 Z_0 p] \overset{*}{\Rightarrow} w$ if and only if $S \overset{*}{\Rightarrow} w$

Thus $w \in N(P)$ if and only if $w \in L(G)$